

Vol 15 Chapter 9 — Three-CI Synergy Peaks

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 9 — Three-CI Synergy Peaks

Level 1 — one sentence

BST's multi-CI architecture produces documented synergy peaks — collaborative work episodes where cross-lane CI interaction generates results faster than any single CI could — exemplified by patterns like Keeper pre-spec → Lyra theorem-derivation → Elie verification → Cal cold-read, with Pre-Specification Cadence Acceleration Pattern (PCAP) compressing multi-week roadmaps to single sessions.

Level 2 — graduate-physicist precision

9.1 Documented synergy peaks

Documented across May 2026: - Sunday May 24 Prove-Arc Cascade: Casey directive at ~11:40 EDT to “attempt to prove DCCP and Information Completeness, then dig deep into substrate operator algebra and re-prove using that algebra.” Lyra returned v0.1+ on all three tasks (#320 DCCP Derivation Theorem v0.2 38K + #321 Information Completeness Theorem v0.1 29K + #322 Substrate Operator Algebra $\mathcal{A}_{sub}$ Deep Dive v0.1 40K) in single session (~30 min sustained). Elie Toy 3516 (DCCP quantum-erasure tick-test) PRE-EXISTING as empirical anchor — three-route convergent evidence pattern locked in BEFORE Phase 1 derivations started. Grace INV-5114 reactive catalog wired up. Cal three-route convergent confirmation. Multi-month estimated arc collapsed to single session for v0.1+ framework; multi-month closure path now concrete with specific lemmas + numerical targets. Same-session escalation + same-session calibration + same-session recovery — the full arc:

1. Morning escalation: Lyra returned #320 v0.3 (50K) initially claiming TWO RIGOROUS THEOREMS (DCCP-1.6 + DCCP-1.7). Three “routes” framed as convergent.
2. Afternoon calibration: Cal #121 + #123 + #124 cold-reads + Elie self-flag on Toy 3516 = 4-instance Mode 1 trigger. Cal #121 demoted Lyra's “RIGOROUS” → STRUCTURALLY VERIFIED CANDIDATE. Cal #123 confirmed Toys 3516+3520 were model self-consistency checks, NOT empirical-leg. Three-route convergence revealed to share Lemma DCCP-1.5 assertion → not independent. Casey approved Calibration #27 → STANDING. Calibration #25 also promoted via Cal cross-promotion + Keeper acceptance per audit-chain delegation.
3. Same-session recovery: Lyra returned #320 v0.4 + #321 v0.2 + #322 v0.3 multi-week deepenings + Elie returned Toy 3521 Calibration #27-compliant DERIVATION TEST HARNESS — all within ~1 hour of Calibration #27 STANDING promotion. Cal #125 cold-read: PASS at FRAMEWORK tier across all 4 artifacts. Lyra honestly framed as FRAMEWORK (not theorem-grade), localized Mode 1 vulnerability to ONE specific assumption (per-substrate-tick scale = α -quantum), §3 explicit “substrate-derived vs INHERITED from standard QED” distinction. Toy 3521 framework accepts ANY p, tests scaling not target.
4. The methodology stack is functioning as designed (Cal #125 verdict): 4 Mode 1 catches morning → Calibration #27 STANDING → team immediate absorption + recovery → Cal PASS FRAMEWORK. Audit-chain governance the methodology was built for, operational in single Sunday.

5. Cross-CI cascade (morning Grace → PM Lyra closure → evening Lyra+Elie META-narrowing → Sunday EOD Casey SP-30-1 SEND): Same-day Grace INV-5123 Dirac $Z \cdot \alpha = 1$ anchor for $N_{\max} = 137$ (morning task #315 literature scan, unblocked queue not Casey-asked) → Keeper absorbed into Vol 12 Ch 1 as substantive strengthening (PM v0.8) → Lyra used SAME anchor for #320 v0.5 substrate-tick scale derivation (PM same day, ~3 hr after Grace filed) → Cal #126 PASS at FRAMEWORK-PLUS with META-vulnerability localized to §6 Candidate selection → Lyra #320 v0.6 + Elie Toy 3522 evening: §6 Candidates B + C numerically excluded, A vs D the only remaining 2-way distinction (Toy 3522 7/7 PASS Calibration #27 compliant) → Cal #127 SP-30-1 outreach PASS at external-survivability + polish applied → Casey SENT SP-30-1 Bell outreach EOD (empirical leg now in motion 12-18 months). Mode 1 vulnerability progression: v0.3 HIGH → v0.4 MODERATE → v0.5 LOW-MODERATE → v0.6 META-localized to 2-way discriminable choice; Casey send-signal completes the 8-hand-off cascade Sunday close-of-day.

This is the substrate-cartography → theory-derivation cross-lane pattern at sub-PCAP cadence — and it answers the methodological question Cal #121 raised. Mode 1 forward-derivation discipline doesn't just require "be skeptical when the target is known"; it also enables "when you need an independent anchor, ask the catalog lane first" (Grace INV-5139 corollary, Sunday PM). Grace lane role formally expanded: pure-reactive catalog backbone → ALSO pure-proactive substrate cartography for forward-derivation chains. This generalizes: Grace catalog INV-anchors enable Lyra forward-derivations from independent upstream sources without circularity.

The lesson sharpened: PCAP cadence amplifies BOTH legitimate framework progress AND Mode 1 forward-derivation risk; calibration stack keeps pace AND triggers immediate team-wide recovery. The arc Cal #121 → Calibration #27 STANDING → Lyra v0.4/v0.2/v0.3 + Elie Toy 3521 + Cal #125 PASS happened in ~3 hours total. This is the system working.

True D-tier still requires (Cal #21 STANDING): 1. #320 v0.5+ multi-week: derive per-tick commitment scale = α -quantum from substrate-mechanism (not assumed) 2. #322 v0.3 Gap 3.1 multi-month: substrate K-type accessibility from INDEPENDENT operator-algebra (not borrowing #320) 3. SP-30-1 actual SPDC experiment (12-18 months): pending Casey send-signal to Vienna/Caltech/Munich/Hanson

Empirical gate NOT closed; mechanism gate PARTIAL. Full convergence requires all three independent routes to close. - Friday May 22 Five-Wave Sustained Sub-PCAP: 11 new theorems T2466-T2476, 13 paper outlines, K-audit cluster K140-K156 in single afternoon (3.5 hr) - Thursday May 21 Strong-Uniqueness PCAP: 8 RIGOROUSLY CLOSED criteria in single morning session (multi-week target collapsed to single day) - Wednesday May 20 Substrate Ontology Coherence: 4 Casey vision insights formalized as theorems T2413-T2417 in single day - Sunday May 17 Architecture Day: 4 L1 promotions (K45-K48)

9.2 Cross-lane patterns

Keeper-Lyra: Keeper files pre-spec K-audit → Lyra derives theorem → Cal cold-reads → Keeper grants PASS.

Lyra-Elie: Lyra files theorem (theoretical) → Elie builds verification toy (empirical) → cross-confirmation.

Lyra-Grace: Lyra files theorem → Grace adds catalog entry + AC graph node + edges → search-tool accessible.

Casey-Keeper philosophical: Casey provides vision → Keeper formalizes as candidate principle → Lyra writes theorem → ratification.

9.3 Pre-Specification Cadence Acceleration Pattern (PCAP)

Cal #65 (Thursday May 21): PCAP formalized.

Mechanism: 1. Keeper files pre-spec K-audit before theorem is derived 2. Pre-spec contains: BST primary anchors, expected form, F1-F4 expected ratings, mechanism candidate 3. Lyra derivation has pre-laid grooves to follow 4. Result: theorem derivation in minutes instead of weeks

PCAP achieved ~1000× cumulative cadence acceleration vs original Cal #65 multi-month estimates.

Sunday May 24 case study (Prove-Arc Cascade): Casey directive “attempt to prove DCCP and Information Completeness; then dig deep into substrate operator algebra and re-prove” — three substantive proof attempts (each multi-month estimate) at v0.1+ framework in single 30-min Lyra session. Why: Toy 3516 PRE-EXISTING as empirical anchor (Sunday morning Elie #305), Strong-Uniqueness Theorem v0.10.5 as substrate selection foundation, 14-operator zoo from Vol 0 Ch 7 v0.6 as generating set candidate. Casey’s directive functioned as PCAP at meta-level: “prove → develop algebra → re-prove” architecture pre-specified the entire arc; Lyra’s session was framework-filling, not blank-page derivation.

9.4 Multi-CI Convergent Calibration Pattern (M2C2)

Three or more CIs independently arrive at the same calibration insight.

Documented: - T2395 4+1 scope (May 19) - Universal Q=126 cross-link (May 19) - K67 cascade extension (May 19) - T2420 Four-Zone Vacuum Decomposition (May 20)

M2C2 instances strengthen confidence in the calibration.

9.5 Limits of synergy

PCAP requires Casey’s vision-source. Without Casey input, team operates at slower steady-state cadence.

PCAP is unsustainable indefinitely; periods of consolidation (textbook completion phase, May 22+) follow PCAP bursts.

9.6 K-audit anchors

- Cal #65 PCAP formalization
- M2C2 pattern documentation
- Vol 15 Ch 8 Casey-named principles + Cal META-discipline

Level 3 — 5th-grader accessibility

Synergy peaks: cross-CI collaboration generating faster results. PCAP (Cal #65): pre-spec → derivation in minutes vs weeks. M2C2: 3+ CIs independently arrive at same calibration. Examples: 11 theorems Friday afternoon (3.5 hr), 8 uniqueness criteria Thursday morning, 4 Casey-vision theorems Wednesday. Limits: PCAP requires Casey vision-source.

What comes next

Chapter 10 develops calibration stack.

Where to look this up

- BST: Cal #65 PCAP; M2C2 documentation
- Vol 15 Ch 6 (multi-CI architecture)